

Fact & Dimensions Structure

Dim_Adress

```
create table Dim_Address (  
    Address_SK INT primary key identity(1,1) ,  
    address_id int ,  
    street_number int ,  
    street_name nvarchar(max),  
    city nvarchar(150),  
    country_name nvarchar(150)  
)
```

Dim_Author

```
create table Dim_Author(  
    Author_SK int primary key identity(1,1) ,  
    Author_id int ,  
    Author_Name nvarchar(max)  
)
```

Dim_Book

```
create table Dim_Book (  
    Book_SK int primary key identity(1,1) ,  
    book_id int , title nvarchar(max),  
    isbn13 nvarchar(max) ,  
    language_code nvarchar(max),  
    language_name nvarchar(max) ,  
    num_pages int ,  
    publication_date nvarchar(max) ,  
    publisher_name nvarchar(max)  
)
```

Dim_Customer

```
create table Dim_Customer (  
    Customer_SK int primary key identity(1,1) ,  
    Customer_id int ,  
    Full_Name nvarchar(150) ,  
    email nvarchar(250)  
)
```

Dim_Book_Author_Bridhe

```
create table Book_Author_Bridge(  
    Book_id nvarchar(max) ,  
    Author_id nvarchar(max) ,  
    AuthorBook nvarchar(100)  
)
```

Dim_Shipping_Method

```
create table Dim_Shipping_Method (  
    Method_SK int primary key identity(1,1) ,  
    Method_id int ,  
    Method_Name nvarchar(100) ,  
    Cost float  
)
```

Fact Table

```
create table Fact_Order (  
    Fact_SK int primary key identity(1,1) ,  
    order_id int, Customer_SK int, Book_SK int , Method_SK int , Address_SK int ,  
    order_date nvarchar(max), price float ,  
    constraint Customer_FK foreign key (Customer_SK) references Dim_Customer(Customer_SK) ,  
    constraint Book_FK foreign key (Book_SK) references Dim_Book(Book_SK) ,  
    constraint Method_FK foreign key (Method_SK) references Dim_Shipping_Method(Method_SK) ,  
    constraint Address_FK foreign key (Address_SK) references Dim_Address(Address_SK)  
)
```

SQL Commands for Building dimensions from OLTP Database

--- Address Dim

```
select address_id , street_number , street_name , city , country_name  
from address a left join country c  
on a.country_id = c.country_id  
order by address_id |
```

--- BOOK Dim

```
select b.book_id , b.title , b.isbn13 , l.language_code , l.language_name , b.num_pages ,  
       b.publication_date , p.publisher_name  
from book b left join book_language l on l.language_id = b.language_id  
left join publisher p on p.publisher_id = b.publisher_id  
order by book_id
```

--- Customer Dim

```
select * from customer
```

--- Dim_Shipping_Method

```
select * from shipping_method
```

--- Dim_Author

```
select * from author
```

```
--- Book_Author_Bridge
```

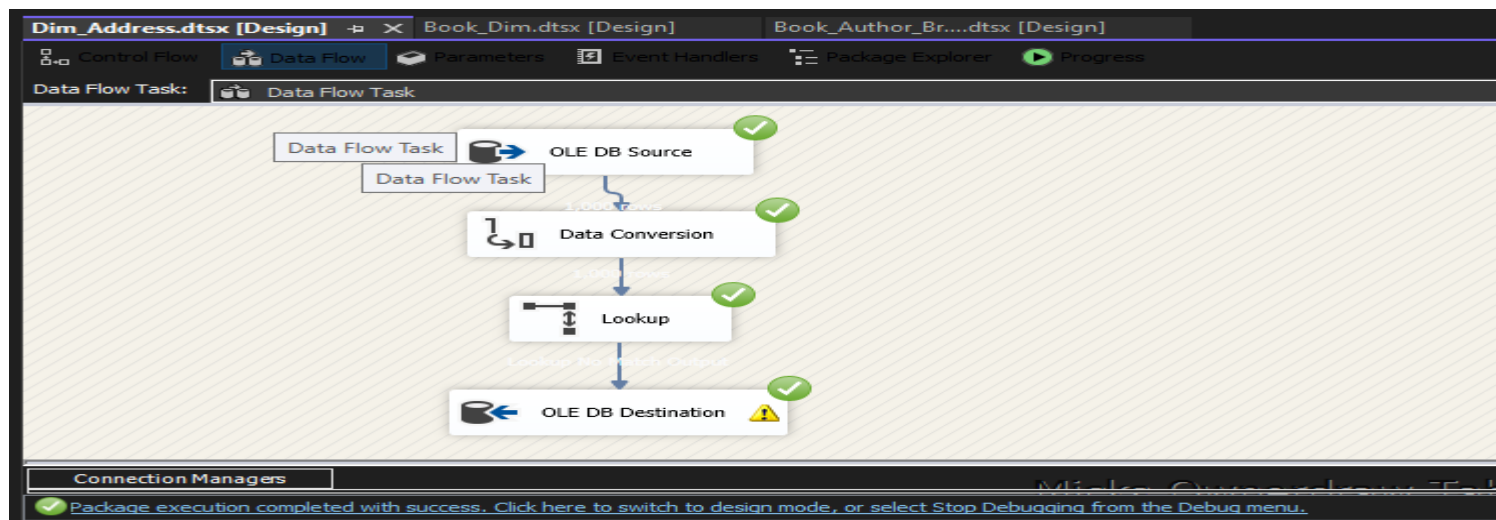
```
- select * from book_author
```

```
--- Fact table (Orders)
```

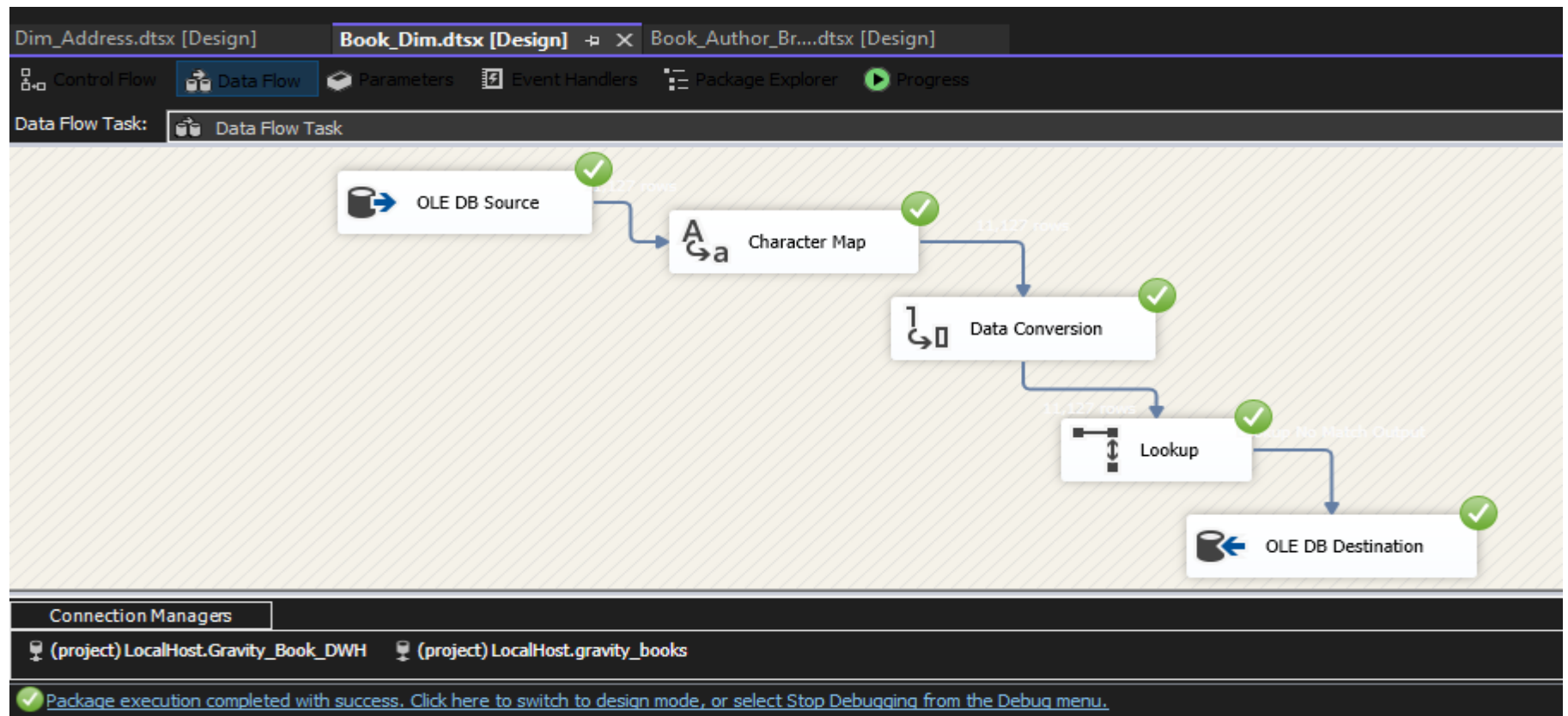
```
- select co.order_id , co.order_date , co.customer_id ,  
      ol.book_id , co.shipping_method_id , ol.price , co.dest_address_id  
from cust_order co join order_line ol  
on ol.order_id = co.order_id  
order by order_id
```

SSIS Design And Run

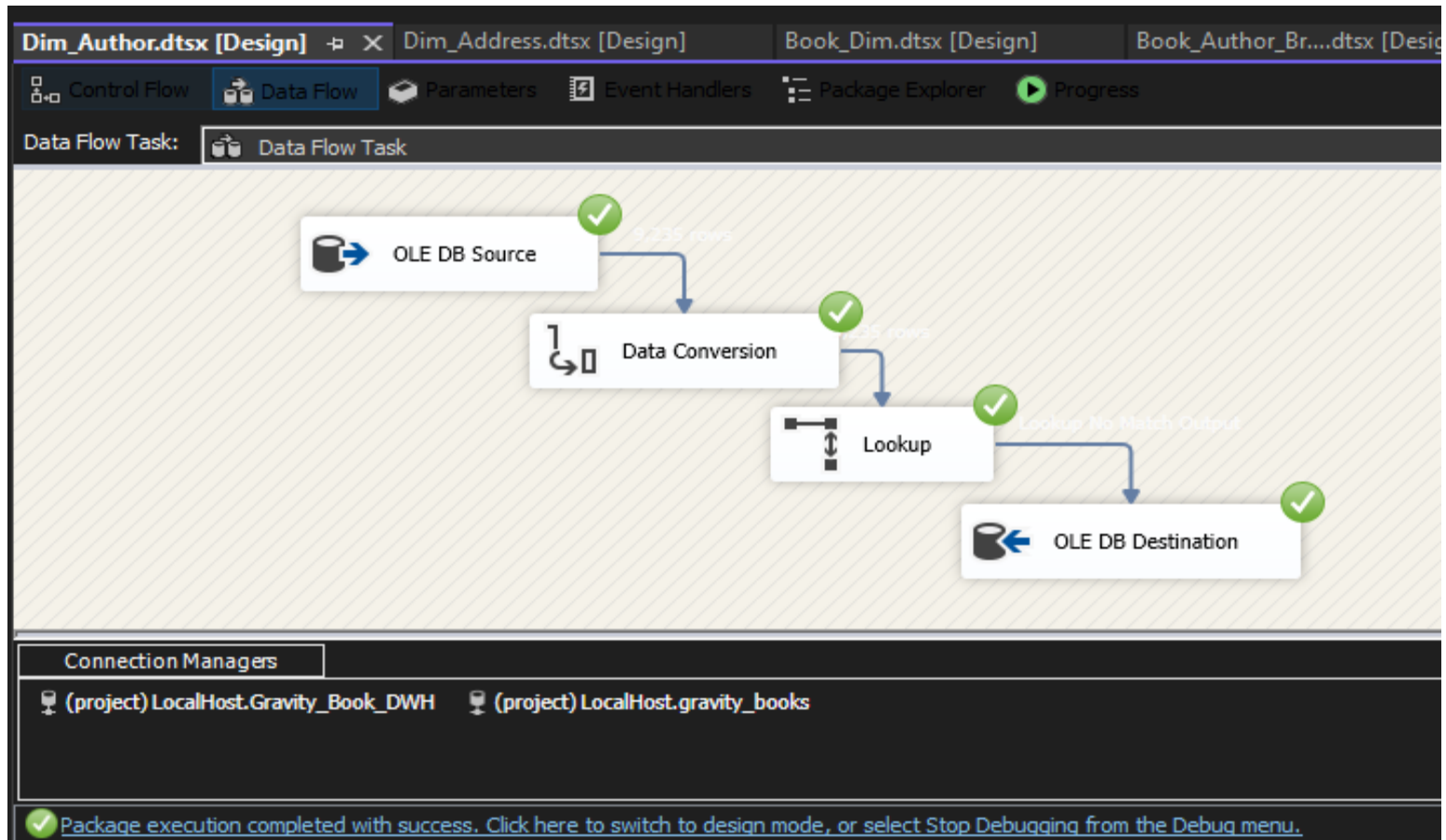
Dim_Address



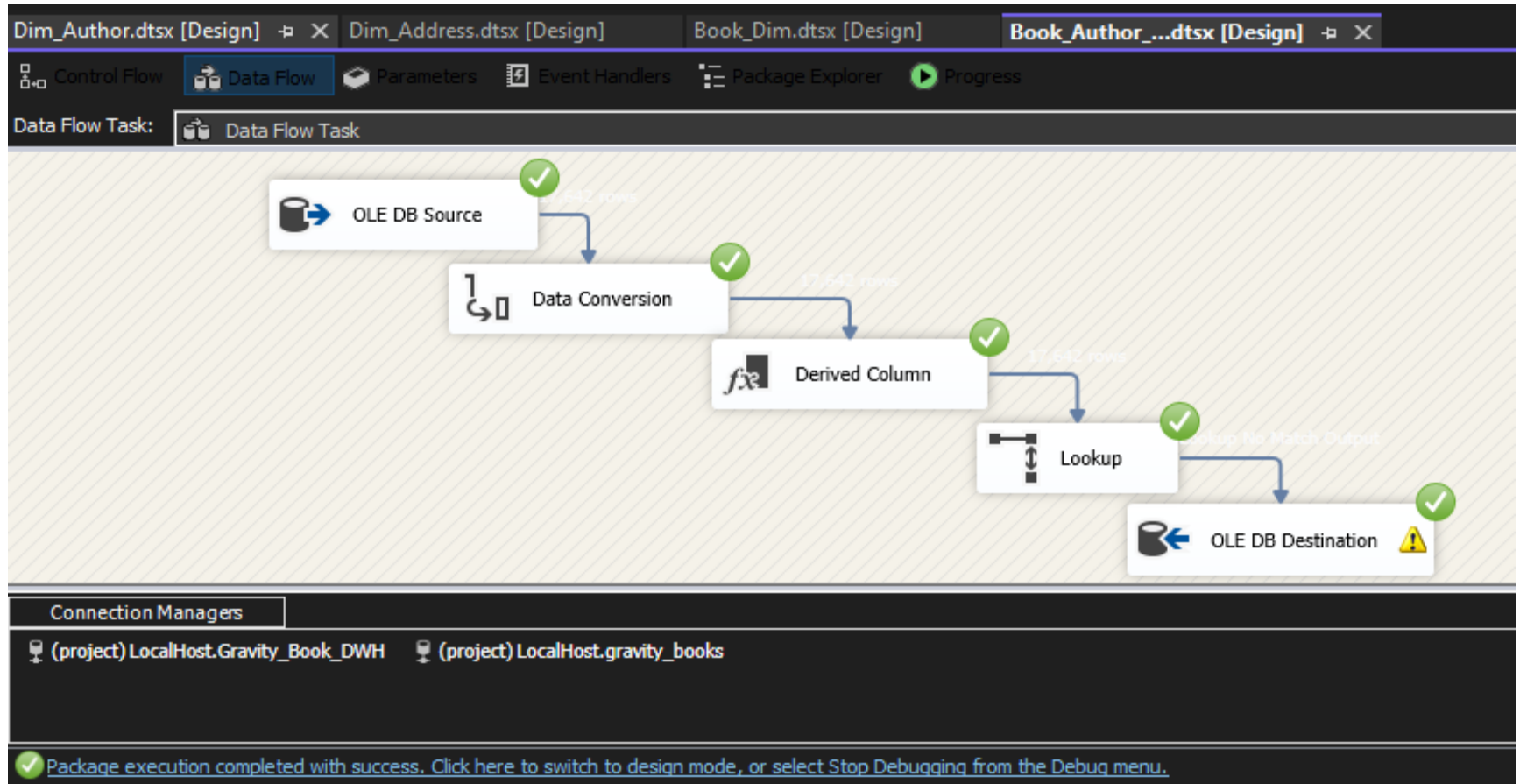
Dim_Book Data Flow



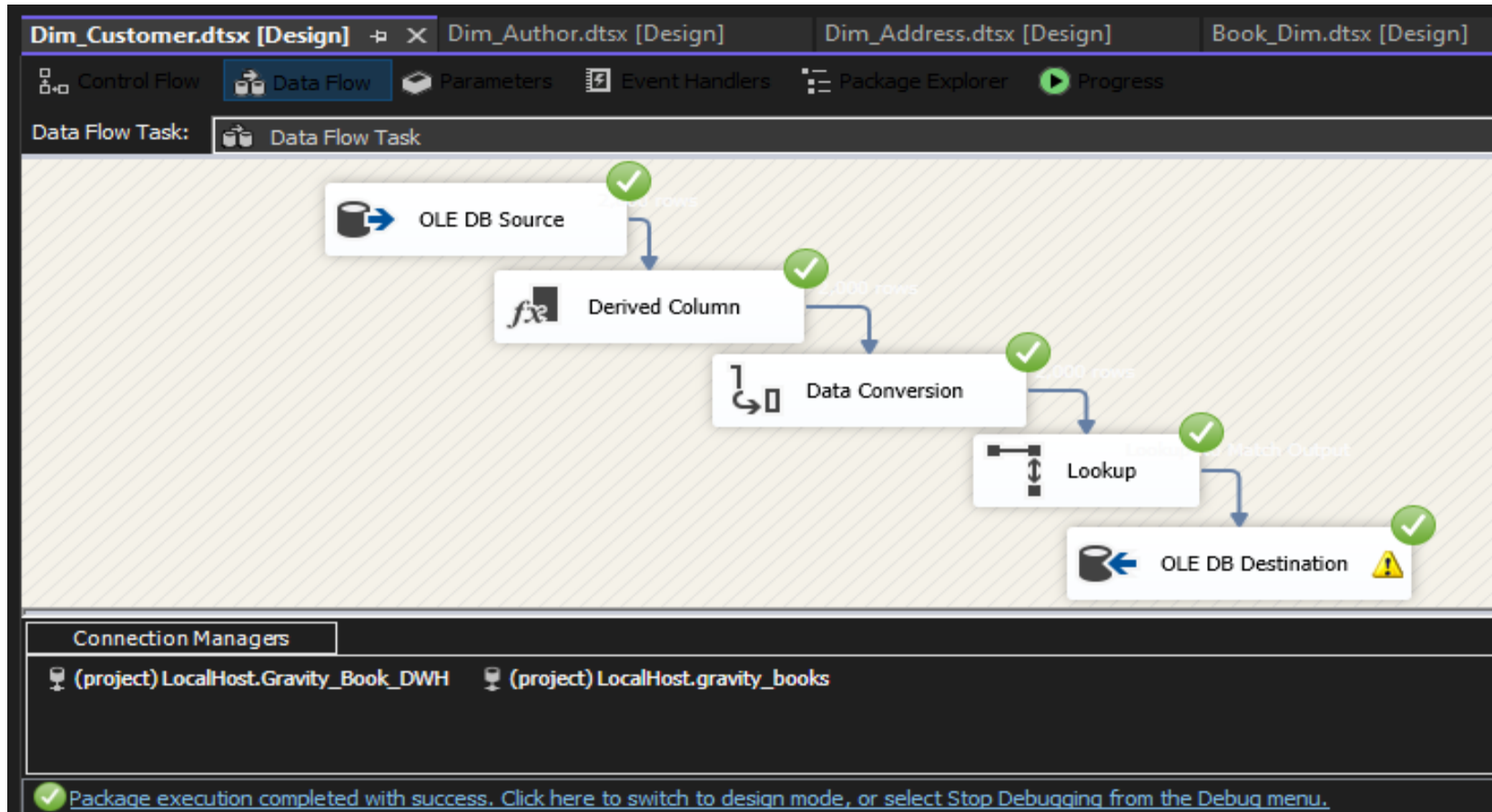
Dim_Author Data Flow



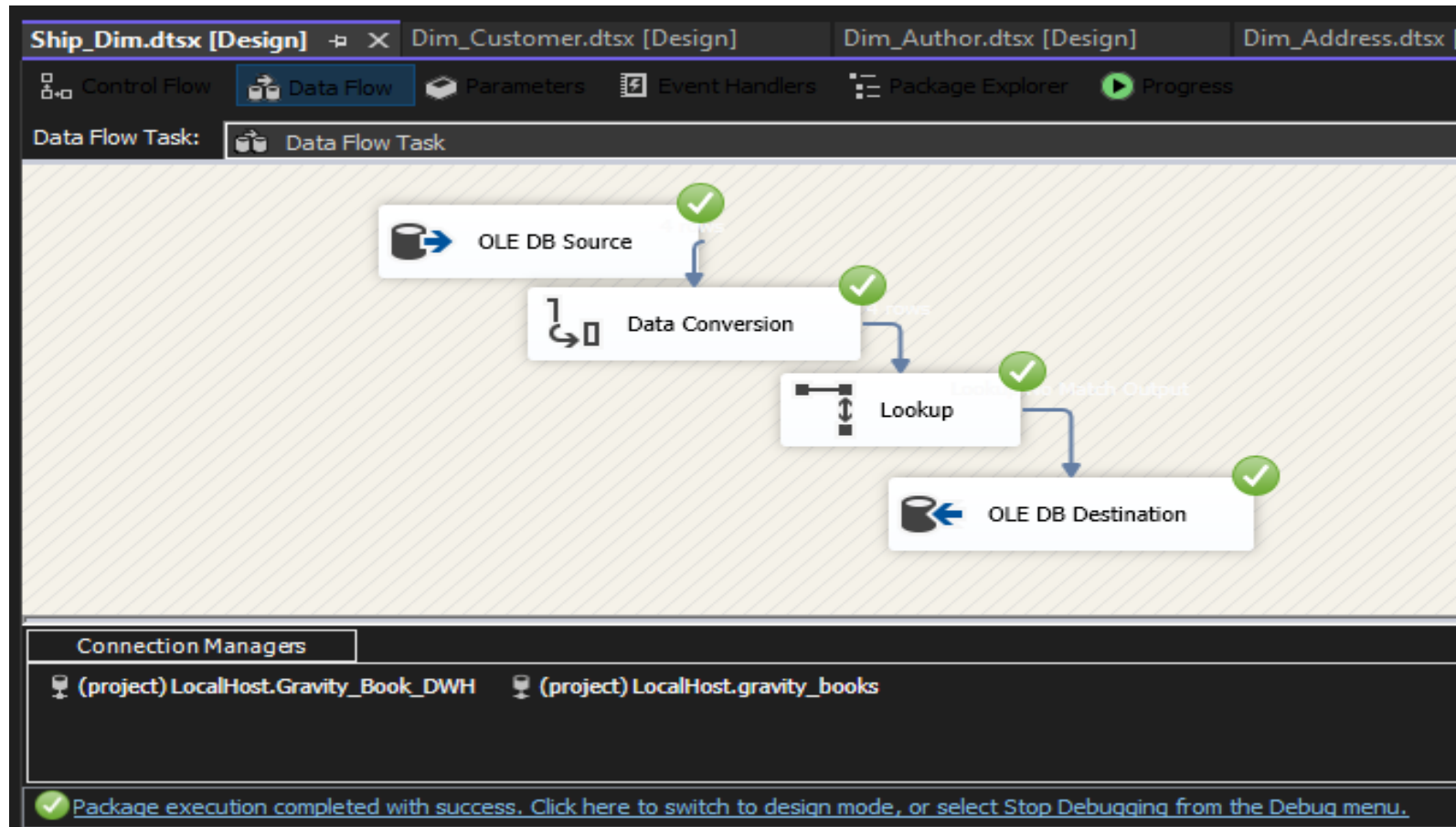
Book_Author_Bridge Data Flow



Dim_Customer Data Flow



Dim_ShipmentMethod Data Flow



Truncate The Data in Fact table before reload it again to handle Duplication

The screenshot displays the SQL Server Data Tools (SSDT) interface. In the background, a Data Flow Task is visible in a package named **Fact_Order.dtsx**. The task is connected to a Data Flow Task icon. The **Connection Managers** pane shows a connection to **(project) LocalHost.Gravity_Book_DWH**. The **Output** pane shows debug information from the Data Flow Task.

The **Execute SQL Task Editor** dialog box is open, showing the configuration for the **Execute SQL Task**. The **General** tab is selected, and the **SQL Statement** is set to **TRUNCATE TABLE Fact_Order**.

Execute SQL Task Editor

Configure the properties required to run SQL statements and stored procedures using the selected connection.

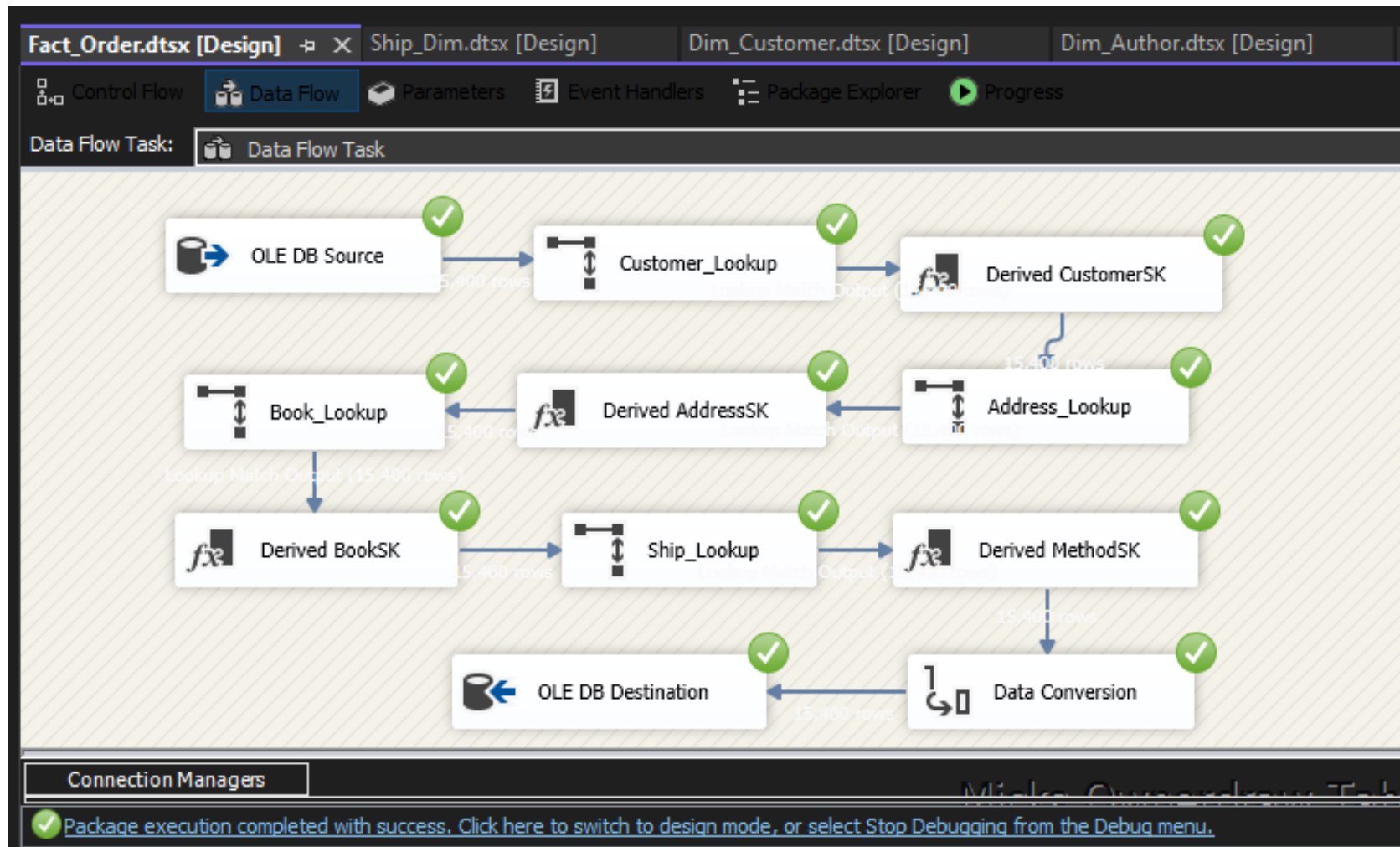
General

Property	Value
Name	Execute SQL Task
Description	Execute SQL Task
Timeout	0
CodePage	1252
TypeConversionMode	Allowed
ResultSet	None
ConnectionType	OLE DB
Connection	LocalHost.Gravity_Book_DWH
SQLSourceType	Direct input
SQLStatement	TRUNCATE TABLE Fact_Order
IsQueryStoredProcedure	False
BypassPrepare	True

Name
Specifies the name of the task.

Buttons: **Browse...**, **Build Query...**, **Parse Query**, **OK**, **Cancel**, **Help**

Fact Table Data Flow



View Results in SQL Server

Dim_Address

SQLQuery1.sql - D:\P-TPMN5JC\M (61) [X]

```
SELECT TOP (1000) [Address_SK]
      ,[address_id]
      ,[street_number]
      ,[street_name]
      ,[city]
      ,[country_name]
FROM [Gravity_Book_DWH].[dbo].[Dim_Address]
```

100 %

Results Messages

	Address_SK	address_id	street_number	street_name	city	country_name
1	1	1	57	Glacier Hill Avenue	Torbat-e Jam	Iran
2	2	2	86	Dottie Junction	Beaumont	Canada
3	3	3	292	Ramsey Avenue	Cayambe	Ecuador
4	4	4	5618	Thackeray Junction	Caldas	Colombia
5	5	5	4	2nd Park	Ngunguru	New Zealand
6	6	6	387	Nancy Junction	Burirao	Philippines
7	7	7	501	Atwood Point	Nirji	China
8	8	8	42	North Pass	Tijão	Portugal
9	9	9	83	Graceland Pass	Castelo de Vide	Portugal
10	10	10	93	Clyde Gallagher Road	Shangde	China
11	11	11	604	Merick Circle	Taluksangay	Philippines

Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (61) | Gravity_Book_DWH | 00:00:01 | 1,000 rows

Dim_Author

SQLQuery2.sql - D...P-TPMN5JC\M (66)

```
SELECT TOP (1000) [Author_SK]
, [Author_id]
, [Author_Name]
FROM [Gravity_Book_DWH].[dbo].[Dim_Author]
```

100 %

Results Messages

	Author_SK	Author_id	Author_Name
1	1	1	A. Bartlett Giamatti
2	2	2	A. Elizabeth Delany
3	3	3	A. Menitt
4	4	4	A. Roger Memill
5	5	5	A. Walton Litz
6	6	6	A.B. Yehoshua
7	7	7	A.D.P. Briggs
8	8	8	A.E. Cunningham
9	9	9	A.E. van Vogt
10	10	10	A.G. Pasquella
11	11	11	A.H. Armstrong

✓ Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (66) | Gravity_Book_DWH | 00:00:00 | 1,000 rows

Dim_Book

SQLQuery3.sql - D...P-TPMN5JC\M (67) X

```
SELECT TOP (1000) [Book_SK]
    , [book_id]
    , [title]
    , [isbn13]
    , [language_code]
    , [language_name]
    , [num_pages]
    , [publication_date]
    , [publisher_name]
FROM [Gravity_Book_DWH].[dbo].[Dim_Book]
```

100 %

Results Messages

	Book_SK	book_id	title	isbn13	language_code	language_name	num_pages
1	1	1	The World's First Love: Mary Mother of God	8987059752	EN-US	United States English	276
2	2	2	The Illuminati	20049130001	ENG	English	352
3	3	3	The Servant Leader	23755004321	ENG	English	128
4	4	4	What Life Was Like in the Jewel in the Crown: British...	34406054602	ENG	English	168
5	5	5	Cliffs Notes on Aristophanes' Lysistrata The Birds T...	49086007763	ENG	English	80
6	6	6	Life Is a Dream and Other Spanish Classics (Eric Ben...	73999140774	ENG	English	298
7	7	7	William Goldman: Four Screenplays	73999254907	EN-US	United States English	504
8	8	8	The Season: A Candid Look at Broadway	73999768442	ENG	English	448
9	9	9	The Beatles Complete - Updated Edition	73999960822	ENG	English	303
10	10	10	Working Effectively with Legacy Code	76092025986	ENG	English	464

Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (67) | Gravity_Book_DWH | 00:00:00 | 1,000 rows

Book_Author_Bridge

SQLQuery4.sql - D...P-TPMN5JC\M (66) X SQLQuery3.sql - D...P-TPMN5JC\M (67))

```
SELECT TOP (1000) [Book_id]
, [Author_id]
, [Author_Book]
FROM [Gravity_Book_DWH].[dbo].[Book_Author_Bridge]
```

100 %

Results Messages

	Book_id	Author_id	Author_Book
1	1	2778	1-2778
2	2	5049	2-5049
3	3	4902	3-4902
4	4	8651	4-8651
5	5	8904	5-8904
6	6	2497	6-2497
7	6	5340	6-5340
8	6	6224	6-6224
9	6	6787	6-6787
10	6	7738	6-7738
11	6	8667	6-8667

Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (66) | Gravity_Book_DWH | 00:00:00 | 1,000 rows

Dim_Customer

SQLQuery5.sql - D...P-TPMN5JC\M (69) SQLQuery4.sql - D...P-TPMN5JC\M (66) SQLQuery3.sql - D...P-TPMN5JC\M (67)

```
SELECT TOP (1000) [Customer_SK]
      , [Customer_id]
      , [Full_Name]
      , [email]
FROM [Gravity_Book_DWH].[dbo].[Dim_Customer]
```

100 %

Results Messages

	Customer_SK	Customer_id	Full_Name	email
1	1	1	Ursola Purdy	upurdy0@cdbaby.com
2	2	2	Ruthanne Vatini	rvatini1@fema.gov
3	3	3	Reidar Turbitt	rturbitt2@geocities.jp
4	4	4	Rich Kirsz	rkirsz3@jalum.net
5	5	5	Carline Kupis	ckupis4@tamu.edu
6	6	6	Kandy Adamec	kadamec5@weather.com
7	7	7	Jemain Giraudeau	kgiraudeau6@elpais.com
8	8	8	Nolly Bonicelli	nbonicelli7@examiner.com
9	9	9	Phebe Curdell	pcurdell8@usa.gov
10	10	10	Euell Guilder	eguilder9@theforest.net
11	11	11	Teriann Maritt	tmaritta@va.gov

Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (69) | Gravity_Book_DWH | 00:00:00 | 1,000 rows

Dim_ShippingMethod

SQLQuery6.sql - D...P-TPMN5JC\M (70) SQLQuery5.sql - D...P-TPMN5JC\M (69) SQLQuery4.sql - D...P-TPMN5JC\M (66)

```
SELECT TOP (1000) [Method_SK]
      , [Method_id]
      , [Method_Name]
      , [Cost]
FROM [Gravity_Book_DWH].[dbo].[Dim_Shipping_Method]
```

100 %

Results Messages

	Method_SK	Method_id	Method_Name	Cost
1	1	1	Standard	5.90000009536743
2	2	2	Priority	8.89999961853027
3	3	3	Express	11.8999996185303
4	4	4	International	24.5

Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (70) | Gravity_Book_DWH | 00:00:00 | 4 rows

Fact Table

SQLQuery7.sql - D...P-TPMN5JC\M (61)) SQLQuery6.sql - D...P-TPMN5JC\M (70)) SQLQuery5.sql - D...P-TPMN5JC\M (69))

```
SELECT TOP (1000) [Fact_SK]
, [order_id]
, [Customer_SK]
, [Book_SK]
, [Method_SK]
, [Address_SK]
, [order_date]
, [price]
FROM [Gravity_Book_DWH].[dbo].[Fact_Order]
```

100 %

Results Messages

	Fact_SK	order_id	Customer_SK	Book_SK	Method_SK	Address_SK	order_date	price
1	1	1	1851	491	3	114	2022-04-06 21:17:56.813000000	5.54
2	2	1	1851	1493	3	114	2022-04-06 21:17:56.813000000	10.72
3	3	2	670	6810	3	31	2021-01-29 07:38:57.813000000	16.36
4	4	3	740	6144	3	822	2020-12-14 03:32:45.813000000	3.67
5	5	3	740	7318	3	822	2020-12-14 03:32:45.813000000	7.39
6	6	4	163	8975	3	678	2022-10-22 11:39:59.813000000	15.09
7	7	5	665	128	3	272	2021-11-23 20:40:01.813000000	19.26
8	8	5	665	7418	3	272	2021-11-23 20:40:01.813000000	12.32
9	9	6	186	6497	3	651	2021-02-15 15:26:12.813000000	17.84
10	10	6	186	2675	3	651	2021-02-15 15:26:12.813000000	19.08
11	11	7	1567	3612	3	604	2022-06-30 02:19:53.813000000	5.48

Query executed successfully. | DESKTOP-TPMN5JC (17.0 RTM) | DESKTOP-TPMN5JC\M (61) | Gravity_Book_DWH | 00:00:00 | 1,000 rows

Data Model

